

CUTE LAYOUT · TENSOR CORE · FLASH ATTENTION

LEETCUDA

CUDA KERNEL 优化之路

从编程模型到 CUTE 与 FLASHATTENTION

→ mode 1: (8)

B = (2,2):(1,8)

0	8	16	24	32	40	48	56	64	72	80	88	96	104	112	120
1	9	17	25	33	41	49	57	65	73	81	89	97	105	113	121
2	10	18	26	34	42	50	58	66	74	82	90	98	106	114	122
3	11	19	27	35	43	51	59	67	75	83	91	99	107	115	123
4	12	20	28	36	44	52	60	68	76	84	92	100	108	116	124
5	13	21	29	37	45	53	61	69	77	85	93	101	109	117	125
6	14	22	30	38	46	54	62	70	78	86	94	102	110	118	126
7	15	23	31	39	47	55	63	71	79	87	95	103	111	119	127

mode 0: (1) ↓

L = (8,16):(1,8)

```
auto tiled_mma = make_tiled_mma(SM80_16x8x16_F32F16F16F32_TN{}, Layout<Shape<2_2_1>>{});
```

郑重声明
本书仅用于开源学习与技术交流，
未经作者本人同意，不得用于任何商业用途。

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李航《统计学习方法》笔记

从原理到实现：基于 R

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

$$\hat{y} = \arg \max_y P(y|x)$$

$$R_{\text{emp}}(f) = \frac{1}{N} \sum_{i=1}^N L(y_i, f(x_i))$$

$$H(P) = -\sum_i p_i \log p_i$$

$$p(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$L(y, z) = \max(0, 1 - yz)$$

$$w \leftarrow w - \eta \nabla L(w)$$

$$H(p, q) = -\sum_i p_i \log q_i$$

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Certificate of Excellence

THIS IS TO CERTIFY THAT

Deftruth

Has successfully contributed to the
Diffusers MVP program.



2026-01-15

Date

Diffusers team at
Hugging Face